

Algebra 1
Unit 1
Lesson 5 Practice

Name _____
Date _____
Period _____

1. In radical form, the expression $x^{\frac{1}{3}}$ is equivalent to _____.

2. In exponential form, the expression $\sqrt{x}$ is equivalent to _____.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-729)^{\frac{1}{3}} =$

4. $(243)^{\frac{1}{5}} =$

5. $(225)^{\frac{1}{2}} =$

6. Simplify the expression using the laws of exponents and then find the value.

$$(-27)^{-\frac{2}{9}} \cdot (-27)^{\frac{5}{9}} =$$

The expression $\frac{(-125)^{\frac{11}{9}}}{(-125)^{\frac{8}{9}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?

8. Rewrite the expression as an equivalent expression.

9. Write the simplified expression in radical form.

10. Evaluate the radical expression.